

Field	Details
Date	04 JULY 2026
Team ID	SWTID-2026-2662
Project Name	AI Powered Debt Relief & Financial Recovery Platform

AI Powered Debt Relief & Financial Recovery Platform

An AI-powered web application that helps borrowers manage debt, analyze financial health, and generate professional settlement/negotiation strategies — built with React.js, FastAPI, SQLite, SQLAlchemy ORM, and Google Gemini API.

Overview

The AI Powered Debt Relief & Financial Recovery Platform simplifies and automates debt management for borrowers. Users can record loan details, analyze their financial health, generate AI-powered negotiation strategies, and receive personalized settlement recommendations — all within a secure, modern digital environment.

Core Features

- AI-Powered Settlement Recommendation – Analyzes loan and income details to generate personalized settlement suggestions and debt stress analysis.
- Intelligent Negotiation Letter Generation – Uses Google Gemini AI to draft professional, lender-specific negotiation emails and settlement letters.
- Financial Health Tracking & Loan Analysis – Dashboard for monthly surplus, EMI ratio, debt stress level, and settlement percentages, with a history of AI-generated negotiations.

Tech Stack

Layer	Technology
Frontend	Streamlit / React.js (Vite)
Backend	FastAPI (Uvicorn)
Database	SQLite with SQLAlchemy ORM
AI Engine	Google Gemini API (gemini-1.5-flash)
Auth	JWT (python-jose) + Werkzeug password hashing
Deployment	Docker, Render / AWS
Version Control	Git & GitHub, GitHub Actions (CI/CD)

Project Structure

```
FinRelief-AI/
├── app/
│   ├── main.py
│   ├── database.py
│   ├── auth.py
│   ├── models/
│   ├── schemas/
│   ├── routers/
│   └── services/
├── streamlit_app/
│   ├── app.py
│   ├── api_client.py
│   ├── components.py
│   ├── styles.css
│   └── pages/
├── requirements.txt
├── runtime.txt
└── .env
```

Getting Started

Prerequisites

- Python 3.11+
- Node.js (for React/Vite frontend, if used)
- Google Gemini API key

Installation

```
# Clone the repository
git clone https://github.com/nakkahemasaisree-cmd/AI-powered-debt-relief-and-
financial-recovery-platform.git
cd AI-powered-debt-relief-and-financial-recovery-platform

# Install backend dependencies
pip install -r requirements.txt

# Configure environment variables
cp .env.example .env
# Add your GOOGLE_API_KEY, SECRET_KEY, DATABASE_URL

# Run the FastAPI backend
uvicorn app.main:app --reload

# Run the Streamlit frontend (in a separate terminal)
streamlit run streamlit_app/app.py
```

Modules

Module	Description
Authentication	Secure registration, login, JWT-based session management
Loan Management	Add, view, and delete loan records
Financial Engine	Computes EMI ratio, debt-to-income ratio, surplus, stress level
Settlement Engine	Rule-based settlement percentage & risk scoring
AI Engine	Gemini-powered negotiation strategy & letter generation (with rule-based fallback)
Dashboard	Financial health visualizations and loan prioritization

Testing

- Functional Testing: Registration/login, loan input validation, AI generation, API connectivity — all passed.
- Performance Testing: Conducted using Apache JMeter; average response time 1.42s, throughput 55 req/sec, error rate 0.30%.

Skills & Domain

Python, FastAPI, SQLite, Generative AI, Responsible AI, LLMs, Prompt Engineering, Gemini AI, API Integration, Cloud Computing

Domain: Google Cloud Generative AI

Team

Name	Role
Hema Sai Sree	Team Lead & Backend Developer
Nithya Sree	Project Coordinator & Frontend/UI Developer
Dharani	AI & Financial Processing Developer
Shalini	Backend Developer & Security

Future Scope

- Real-time banking & credit bureau API integration
- Migration to PostgreSQL/MySQL with cloud deployment
- Voice-enabled chatbot & multilingual negotiation support
- Multi-Factor Authentication (MFA) and role-based access control

Disclaimer

This project is an academic/internship proof-of-concept. It does not connect to real banking systems and should not be used as a substitute for professional financial or legal advice.

Links

- GitHub Repository: <https://github.com/nakkahemasaisree-cmd/AI-powered-debt-relief-and-financial-recovery-platform>
- Demo Video: <https://drive.google.com/file/d/1AoQy2YSE4mAHeExVLr-KgtKG2Yc36TBm/view?usp=sharing>